

Vulnerable pedestrian detection and tracking using deep learning

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Abstract

Accidents related with vulnerable pedestrians around crosswalks are continued so that proactive safety support system is required. Our research includes vulnerable pedestrian action character deduction, pedestrian action analysis module development and safety system development using big data analysis for the proactive safety support system.

This paper shows pedestrian/car detection, tracking and action recognition system using deep learning using video streams which come from CCTVs installed at SNU(Seoul National University). This algorithm includes SSD(Single shot multibox detector) structure and mobilenets for faster process and higher detection ratio.

Keywords: Pedestrian detection, Deep learning, Single shot multibox detector, mobilenet

1. Introduction

50.6% of 1,795 dead pedestrians by traffic accident were aged people at 2015 in Korea. The result shows that the ratio is 6 times higher than 13~64 years old and 19.6 times than under 12 years old. The analysis of accident characters and environment can help to reduce accidents. Young children have regularized temporal routines like going school and spatial characters as limited places to play. However, the behavior character analysis of the olds has the limitation in prediction of their action. For these reasons using ICT technology, the analysis technology of vulnerable pedestrian character including the olds is required [1].

In these days, hardware improvement enables us to use deep learning methods. After 2012 in ILSVRC(ImageNet Large Scale Visual Recognition Competition), higher ranked teams used deep learning methods and ResNet from Microsoft had 3.54% error ratio which was close to human

recognition ratio. CNN(Convolutional neural network) based face recognition model of Facebook had 97.25% of correction ratio.

In this paper, SSD[2] and mobilenet[3] which is a kind of CNN are used to detect pedestrians. In chapter 2, our prior algorithm is shown, the video analysis system and results are shown in chapter 3 and 4.

2. Prior research

Real time system for multi-channel pedestrian detection was very tough topic except using GPU(Graphic processing unit) that enabled dramatic increases in computing performance. HoG(Histogram of Gradient), MoG(Mixture of Gaussian) and modified meanshift method were used for multi-channel analysis.

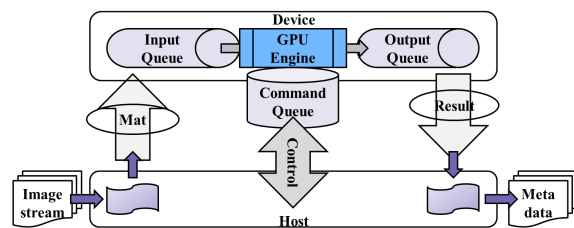


Fig 1: Acceleration blocks for image analysis[4]

This algorithm had fast processing time on GPU, but it showed lower detection ratio comparing with CNN.

3. Video analysis system

3.1 Deep learning structure

Fig 2. shows the structure of SSD which adds several feature layers to the end of base network, which predicts the offset to default boxes of different scales and aspect ratios[2]. SSD used VGG-16 which was very deep and computationally heavy[5]. For

faster process, mobilenet is used as the base network in this study. Mobilenet is based on depthwise separable convolutions which is a form of factorized convolutions[3].

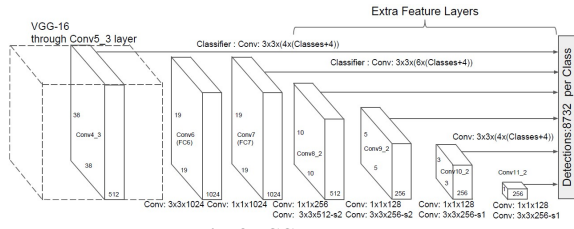


Fig 2: SSD structure

The output feature map of mobilenet is computed as below.

$$G_{k,j,n} = \sum_{i,j,m} K_{i,j,m,n} \cdot F_{k+1,-1,i+j-1,m} \quad (1)$$

Standard convolutions have the computational cost of

$$G_{cost} = D_K \cdot D_K \cdot M \cdot N \cdot D_F \cdot D_F \quad (2)$$

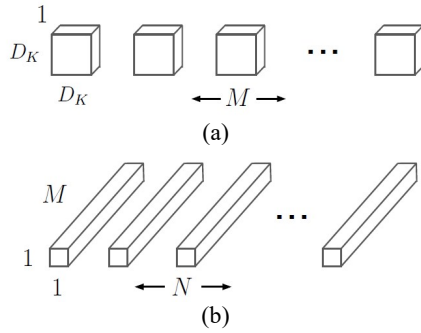


Fig 3: Depthwise convolution

Depthwise convolution has reduced calculation by $ratio = \frac{1}{N} + \frac{1}{D_K^2}$ (3)

Fig 3. Shows depthwise convolution in (a) and pointwise convolution in (b) to build a depthwise separable filter[3].

3.2 Dataset



Fig 4: Recording system installed in SNU

CCTVs were installed in SNU for training and testing in real environment shown in Fig.4. CCTVs are connected on internet and videos have transferred to DVR(Digital Video Recorder) and image analysis server in KETI(Korea Electronics Technology Institute) for 3 months.

4. Experimental results

Fig 5. shows detection results using the trained model acquired from COCO dataset[6] and SNU CCTV dataset. The result shows that for higher detection ratio, real dataset which is captured in real environment is recommended for training.

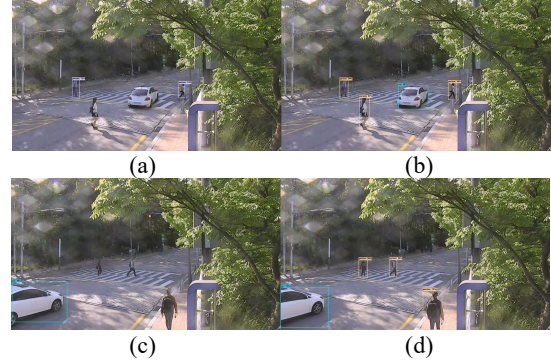


Fig 5: Detection results using trained model with COCO dataset(a,c) and our dataset(b,d)

Table 1: Processing time

Base model	Time (msec)
VGG16	55
MobileNet	12

Table 1. shows processing time results of base models. mobileNet has faster processing time.

5. Acknowledgement

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